

Final Year Project

Creating 360-degree Image Using Mobile Cameras



By

Sakhawat Ali

058950

Mehboob Arshad

058961

Under the supervision of

EBRYX TEAM & MR. MUHAMMAD NAEEM AKHTAR

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY (2021-2025)

AKHUWAT COLLEGE KASUR

AFFILIATED WITH

UNIVERSITY OF THE PUNJAB, LAHORE.

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Signature: -----
Sakhawat Ali 058950

Signature: -----
Mehboob Arshad 058961

CERTIFICATE OF APPROVAL

It is to certify that the final year project (FYP) of BSIT “**Creating 360-degree Image Using Mobile Cameras**” was developed by Sakhawat Ali (058950) and Mehboob Arshad (058961) under the supervision of “Mr. Muhammad Naeem Akhtar”. In my opinion, it is fully adequate, in scope and quality for the degree of Bachelors of Science in Information Technology.

Signature: -----
FYP Supervisor:

Signature: -----
Head of FYP Coordination Office:

Signature:-----
Head of Department of Information Technology

Dated:_____

ACKNOWLEDGEMENT

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ABSTRACT

The project “Creating 360-degree Image Using Mobile Cameras” is designed to provide an accessible and cost-effective solution for capturing and viewing immersive 360-degree environments. Traditional 2D images and videos are limited in representing complex scenes, as they only display a single perspective at a time and often result in distorted or incomplete panoramic views. While specialized 3D cameras exist to overcome this limitation, they are expensive and impractical for small businesses and individual creators.

This system leverages the growing capabilities of modern smartphones, including advanced cameras and image processing features, to bridge the gap between expensive hardware-based solutions and the demand for realistic 360-degree visualization. By enabling multiple Android devices to collaborate in capturing and processing images, the computational load is distributed, enhancing the quality and accuracy of the resulting panoramic view.

The proposed mobile-based solution enables users to create and explore realistic 360-degree scenes anytime and anywhere, supporting applications in tourism, real estate, education, and virtual tours. This approach democratizes immersive visualization technology by offering an affordable, smartphone-driven alternative that empowers both individuals and small businesses.

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1. Introduction

The increasing demand for immersive 360-degree visual experiences is transforming industries such as tourism, real estate, education, and virtual tours. While 2D images and videos offer limited perspectives, they fall short in delivering a fully realistic representation of an environment. Although hardware-based 3D cameras provide high-quality results, their cost makes them impractical for small businesses and individual creators. With the rapid advancement of smartphone capabilities in imaging and processing, there is a clear opportunity to develop an affordable alternative. This project presents a mobile-based solution that utilizes multiple Android devices to collaboratively capture and process images, generating high quality 360-degree views. By distributing the computational tasks across devices, the proposed system offers a cost-effective and accessible method for creating immersive environments without relying on expensive hardware.

1.1 Project Title

“Creating 360-degree image using mobile cameras”

1.2 Project Overview Statement

The trend to capture and view immersive 360-degree environment is growing up in the fields including tourism, real estate, education and virtual tours. These applications demand realistic 360-degree view of environment so that user can explore the environment from anywhere and anytime.

However, 2D images and videos are used to display the 360-degree view partially but these are incapable to give the realistic 360-degree view of environment. There also exist hardware-based camera solution to create 3D views but these are expensive solutions for small business and individual creator.

As smartphones are growing with their cameras and image processing features along with other capabilities so, by leveraging these ones we can come up with cost effective solution which can bridge the gap between expensive solutions and need for 360-degree visualization.

Traditional 2D images are limited in their ability to represent complex 360-degree scenes, as they can only display a single perspective at any given moment. To overcome this limitation, multiple images from various angles are typically captured and combined. However, simply stitching these images into a flat picture often results in a distorted or incomplete view, failing to provide a realistic representation of the entire scene.

While specialized 3D cameras exist to address this challenge, they are expensive, making them inaccessible for most users. This creates a gap for individuals or small businesses that need to generate immersive 3D scenes for applications such as virtual tours, real estate, tourism, and educational purposes.

Therefore, a cost-effective and accessible solution using readily available smartphones is needed. By enabling multiple devices to collaborate in capturing and processing images, it is possible to distribute computational load and enhance the quality of the resulting panoramic view. This project proposes a mobile-based solution that will help to create and view 360-degree image by leveraging distributed image processing across Android devices.

1.3 Project Overview Statement Template

Project Title: Creating 360-degree image using mobile cameras			
Group Leader: Sakhawat Ali			
Project Members:			
Name	Roll no. #	Email Address	Signature
Sakhawat Ali	21F-BSIT-09	sakhawat.raza890@gmail.com	
Mehboob Arshad	21F-BSIT-30	mehboob14@proton.me	
Project Goal: This project aims to develop an android application that allows users to create and view 360-degree image using single as well as multiple smartphones.			
Objectives:			
Sr. No.			
1	Develop a mobile-based solution for capturing 360-degree images using multiple devices.		
2	Develop a system where multiple Android devices are positioned in a circular manner to collectively capture images for a complete 360-degree view.		
3	Implement distributed processing across participating Android devices to balance the computational load, where each device contributes to image stitching and alignment.		
4	Develop single-device mode which enable user to capture images of all sides using a single device and seamlessly stich them together to create 360-degree image.		
5	Guide the user to align and rotate the mobile while taking images using single mobile.		
6	Integrate a 360-degree viewer that allows users to view the final panoramic image in interactively manner, enhancing the user experience.		

7	Design a guide to assist users in arranging devices in a circle and capturing images with minimal distortion and alignment errors.
8	Utilize computer vision techniques for seamless image stitching and enhancement.
Project Success criteria: The project will have following features to considered it as complete. • A functional prototype enabling collaborative image capturing, processing and creating final 360-degree image. • Images should be stitched in a way that there should not be any distortion between images while displaying these as 360-degree image. • Successful integration of 360-degree image viewer. • Implement user guide to take images of required criteria. • Implement seamless inter-device communication and image sharing in multi-device mode. • Create standardize documentation of project.	
Assumptions, Risks, and Obstacles: 1- Integration free availability of 360-degree image viewer. 2- The system may get slow in case of too high resolution images and low hardware capacity.	
Organization Address : 2nd Floor,443-q,phase-ii,dha Lahore (Ebryx)	
Type of project: ☐ Development	
Target End users: 1- Real estate professionals 2- Tourism & travel industry 3- Architects & interior designers 4- Event planners & venues 5- Construction & engineering firms 6- Museums & cultural institutions	
Development Technology: ☐ Object Oriented	
Platform: ☐ Android application	
Suggested Project Supervisor: Mr. Muhammad Naeem Akhtar	
Approved By: Mr. Muhammad Naeem Akhtar	

Date:

1.4 Project Goals & Objectives

Goal

This project aims to develop an android application that allows users to create 360-degree image using single as well as multiple smartphones.

Objectives

- Design a guide to assist users in arranging devices in a circle and capturing images with minimal distortion and alignment errors.
- Develop a system where multiple Android devices are positioned in a circular manner to collectively capture images for a complete 360-degree view.
- Implement distributed processing across participating Android devices to balance the computational load, where each device contributes to image stitching and alignment.
- Develop single-device mode which enable user to capture images of all sides using a single device and seamlessly stich them together to create 360-degree image.
- Integrate a 360-degree viewer that allows users to explore the panoramic image interactively, enhancing the user experience.

1.5 High-level system components

Generally, an integrated application is composed of many different components that work collectively to address the problem for which the system has been devised. The high-level system components / modules for our application are the following:

- Storage & Export Module – Manages local storage of captured/processed images and optional cloud upload for sharing.
- Single device mode
- User Interface (UI) Module
- Synchronization & Coordination Module
- Multi-Device Communication Module
- Image Processing Module
- User Guide & Settings Module
- Multi-device communication

- Camera Module – Captures images from multiple devices using

1.6 List of optional functional units

- Creating a server for image processing
- Develop application for IOS device
- Communication between IOS and Android devices

1.7 Exclusions

- We will not include cloud-based image stitching in the initial phase, as the primary focus is on mobile-based distributed processing.
- The project does not aim to support professional-grade DSLR camera integration

1.8 Application Architecture

We will follow 1-tier Architecture. where all processing and presentation logic reside within the mobile application.

- Kotlin & Jetpack compose (for business & presentation layer)

1.9 Gantt chart

Sr. No.	Task Name	Start Date	End Date	Duration (weeks)
1	Project Proposal	17-10-2024	31-10-2024	2
2	Documentation	01-11-2024	28-11-2024	4
3	Prototype	29-11-2024	05-12-2024	1
4	UI development	06-12-2024	26-12-2024	3
5	Image processing 1	27-12-2025	06-02-2025	6
6	Single device mode	07-02-2025	13-03-2025	5
7	Image processing 2	14-03-2025	24-04-2025	6
8	Multi-device mode	25-04-2025	15-05-2025	3
9	Testing	16-05-2025	22-05-2025	1

10	Deployment	23-05-2025	29-05-2025	1
11	Maintenance	30-05-2025	7-06-2025	1

1.10 Hardware and Software Specification

Hardware: (Recommended)

- 3.6 GHz Octa-core Processor
- 4 GB RAM
- 50 GB Storage
- 12 MP Camera (or higher)
- Gyroscope and Accelerometer

Software: (Recommended)

- Android 12 (API 35)

Table 1 Gantt chart

1.11 Tools and technologies used with reasoning

Tools:

- **Android Studio**
Android Studio is the official IDE for Android development, providing robust tools for building, debugging, and testing applications. It offers features like code analysis, an advanced emulator, and direct integration with Jetpack Compose.
- **OpenCV**
OpenCV (Open Source Computer Vision Library) is used for image processing and computer vision tasks. It provides powerful features for real-time image and video analysis, feature detection, image stitching, and object tracking.
- **Git**
Git is a distributed version control system used for tracking changes in source code during software development. It enables collaboration, branching, and version management, allowing multiple developers to work on a project efficiently without conflicts.

- GitHub

GitHub is a cloud-based platform that provides Git repository hosting with additional features like issue tracking, pull requests, CI/CD integration, and collaboration tools. It enhances project management and enables teams to maintain code quality through version control and code reviews.

Technologies:

- Kotlin

Kotlin is the official programming language for Android development, offering concise syntax, null safety, and interoperability with Java. It enhances developer productivity and ensures modern app development practices.

Frameworks & Libraries

- Jetpack Compose

Jetpack Compose is a modern UI toolkit for building declarative and efficient UI in Android apps. It simplifies UI development with composable functions, state management, and better performance compared to traditional XML-based UI.

- OpenCV (Image Processing)
- CameraX (Android Camera API)
- Communication: Bluetooth/Wi-Fi Direct for device-to-device image transfer
- Storage: Local storage with optional cloud integration

2. Requirements Analysis

The purpose of this document is to outline the functional and non-functional requirements for the **Vision360** system. This system aims to simplify the process of capturing 360 Images using phones Cameras. The SRS document will serve as a guide for the development team to ensure that the system meets the needs of its user.

2.1 User classes and characteristics

The system will have mainly following three types of user.

User Class	User Characteristics
Server user	This is the user of server device, a device which will be used to send the command to client device to capture image. The server user will have the rights to manage the client devices like initiate image capturing process and view the final result.
Client user	Client user is the user of client device, a device which will be connected with server device and instructed by server device to capture and send the image back to sender device. Client user has to allow the connection with server device and align the participant devices in a circular manner to capture a 360-degree scene.
User	It is the user which is using the application. It can handle all the operations like image view, delete and rename image etc.

Table 2 User classes

2.2 Requirement Identifying Technique

The requirements for this system are identified using **Use Case Analysis**. This technique is suitable for interactive end-user applications like **Vision360** as it helps define user interactions and system behavior.

2.2.1 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. The requirements are categorized in different modules.

Authentication System

- User shall be able to register using their email and password.

- User shall be able to log in using the email and password.
- User shall be able to reset their password via email verification.

Single Device Mode

- User shall be able to capture the image.
- System shall guide user to capture next image.
- User shall be able to save the final image.

Multi Device Mode

- User shall be able to select the device as “server” or “client” device.
- Server user shall be able to assign sequence numbering to each device
- Master user shall be able to set the timer for image capturing
- Master user shall be able to initiate image capturing process

Viewer Module

- User shall be able to view the final 360-degree image in an interactive manner
- User shall be able to view the guide for single device Image capturing
- User shall be able to view the guide for multiple device image capturing

Home Module

- User shall be able to search a previously captured image using name.
- User shall be able to rename the captured image.
- User shall be able to delete an image.
- User shall be able to share the image.

Identifier	FR-1
Title	User Authentication
Requirement	User shall be able to register, log in, and reset passwords using email verification. The system shall send a verification email upon registration.
Source	Business Requirement, Security Considerations
Rationale	Ensures secure access to the platform and....
Business Rule (if required)	Allow the user to access the saved images from any device where the application is installed.

Dependencies	None
Priority	High

Identifier	FR-2
Title	Single Device Mode
Requirement	System shall be able to capture, save, display image System shall be able to calculate desired area left, right, up, down which needs to be captured net System shall be able to save or cancel the resultant final image.
Source	Business Requirement, User Needs
Rationale	Provides user an interface to capture seamless calculated images to create quality 360-degree image.
Business Rule (if required)	User must follow the guided steps to capture a full 360-degree image.
Dependencies	FR-1
Priority	High

Identifier	FR-3
Title	Multiple Device Mode
Requirement	System shall be able to connect multiple phones one act as server device System shall confirm the connection with participant devices System shall be able to simultaneously capture images on all devices
Source	User Requirement
Rationale	Helps user to capture 360 seamless image using multiple phones cameras.

Business Rule (if required)	All participating devices must confirm their positions before initiating the capture. A defined numbering sequence is mandatory for efficient image stitching.
Dependencies	Devices must be authenticated to capture collaborative images
Priority	High

Identifier	FR-6
Title	Viewer Module
Requirement	User shall be able to view the final 360-degree images in the scrollable and interactive manner User shall be able to scroll left, right, up, down
Source	Business Requirement, User Experiences
Rationale	Enables user to view, and explore 360 view
Business Rule (if required)	Third party viewer will be integrated
Dependencies	FR-1 (Authentication), FR-3
Priority	Low

Identifier	FR-7
Title	Home Module
Requirement	User shall be able to view the previous captured images, can rename, delete images and share images.
Source	Business Requirement, User Experiences
Rationale	Enables user to view, rename, delete images
Business Rule (if required)	System shall provide mechanism to share images
Dependencies	FR-1 (Authentication), FR-3
Priority	Medium

2.2.2 Non-Functional Requirements

The Vision360 system must meet following quality attributes to ensure reliability, usability, and security:

Reliability

- In case of a device failure during multi-device capture, the system should automatically retry or notify the user to reattempt the session.

Usability

- Guide for Both single and multiple devices should be concise and easy to understand.
- Image capturing and stitching workflows shall be streamlined to require no more than 3 user actions per step.
- The architecture must allow easy integration of cloud-based processing for future scalability.

Performance

- The system shall process and stitch images within 5 seconds for single-device mode and within 10 seconds for multi-device mode.
- Image loading and rendering must be completed within 2 seconds.
- The system shall support a minimum of 5 participating devices

Security

- Error messages shall provide clear user guidance.
- Secure communication protocols HTTPS, Web Sockets with SSL/TLS must be used for data transmission.
- All user credentials must be encrypted using latest encryption algorithms AES.

2.2.3 External Interface Requirements

This section provides information to ensure that the system will communicate properly with user and with external hardware or software elements.

User Interfaces Requirements

- The system shall follow modern **GUI** design standards for accessibility and usability.
- Font styles, button labels, icons, and color schemes shall be consistent across all screens.
- Common navigation links shall be available on every screen.

- The interface shall be responsive and work on different screen sizes.
- Error messages shall provide clear user guidance.

Software interfaces

- The system shall integrate with the OpenCV library for image processing and stitching.
- The system shall use WIFI Hotspot for real-time communication protocol for device-to-device interaction.
- The system shall use Firebase Authentication for user registration, login, and account management.

Hardware interfaces

- The system shall utilize the device's gyroscope and accelerometer to assist in movement tracking and alignment.
- The system shall require Wi-Fi Hotspot for multi-device communication.
- The system shall use internal storage for temporarily storing captured images before processing and final result.

Communications interfaces

- The system shall support data sharing using WIFI hotspot.
- The system shall use Wi-Fi Hotspot for high-speed communication in multi-device mode.

2.3 Use case Analysis

Use Case #1 - Register

<i>UC Identifier</i>	<i>UC-1</i>
<i>Purpose</i>	To allow user to create an account on the platform.
<i>Preconditions</i>	The user is not already registered.
<i>Post conditions</i>	A new account is created.
<i>Actors</i>	Platform User
<i>Extends</i>	None
<i>Main Success Scenario</i>	• The user navigates to the Register page. • The user enter email, and select password. • The system validates the email and password. • The system creates a new account. • The system sends a confirmation email.
<i>Alternate Flows</i>	If the email is already registered, the system displays an error: "Email already in use".

Exceptions	On invalid credentials user got prompted “password atleast 8 characters long”
Includes	None

Table 3: UC-1

Use Case #2 - Login

UC Identifier	UC-2
Purpose	To allow the user to access previously created images from any android device.
Preconditions	The user must have a registered account.
Post conditions	The user is logged into the system.
Actors	User, Server User
Extends	None
Main Success Scenario	• The user navigates to the login page. • User enter the email and password. • The system validates the credentials. • The system grants access to the respective dashboard.
Alternate Flows	If the credentials are invalid, the system displays an error: “Invalid email or password”.
Exceptions	The system fails to validate the credentials due to a server error.
Includes	Register

Table 4: UC-2

Use Case #3 – Forgot Password

UC Identifier	UC-3
Purpose	To enable user to reset their password.
Preconditions	The user must have a registered account.
Post conditions	The user's password is reset.
Actors	User
Extends	None
Main Success Scenario	• The user navigates to profile section. • The user clicks on the "Forgot Password". • User enter the registered email address. • The system sends a password reset link to the email. • The user clicks the link and enters a new password. • The system updates the password.
Alternate Flows	If the email is not registered, the system displays an error: “Email not found”.
Exceptions	The system fails to send the password reset link.
Includes	Register

Table 5: UC-3

Use Case #4 – Capture Image

<i>UC Identifier</i>	UC-4
<i>Purpose</i>	To enable user to Capture an Image.
<i>Preconditions</i>	The user must have a registered account.
<i>Post conditions</i>	An Image is captured, stored and will be used for stitching.
<i>Actors</i>	Server user
<i>Extends</i>	None
<i>Main Success Scenario</i>	• User press the capture button to capture the image only if it is first image. • Otherwise, align the device according to the guide displayed on the screen, repeat step 1.
<i>Alternate Flows</i>	Capture button will remain disabled until desired dot will not be placed inside circle.
<i>Exceptions</i>	The system fails to calculate desired capturing area due to any mobile sensor problem.
<i>Includes</i>	Login

Table 6: UC-4

Use Case #5 – Guide while Capturing

<i>UC Identifier</i>	UC-5
<i>Purpose</i>	To enable user to Capture 360 Images.
<i>Preconditions</i>	The user must have a registered account.
<i>Post conditions</i>	An Image is captured, stored and will be displayed on the screen.
<i>Actors</i>	Platform User(single Device Mode)
<i>Extends</i>	None
<i>Main Success Scenario</i>	• The User need to fit the dot inside circle using sensors. • The system capture the image and shall display on the captured area. • The system calculates the next desired area which needs to be captured and show respective circles. • Again, above 3 steps will repeat until 360-degree will be covered.
<i>Alternate Flows</i>	Capture button will remain disabled until desired dot will not be placed inside circle.
<i>Exceptions</i>	The system fails to calculate desired capturing area due to any mobile sensor problem.
<i>Includes</i>	Login

Table 7: UC-5

Use Case #6 – View Image

<i>UC Identifier</i>	UC-6
<i>Purpose</i>	To enable user to view resultant 360-degree image in an interactive way.
<i>Preconditions</i>	The user must have an captured image.
<i>Post conditions</i>	Stored Image will be displayed on the screen and user can scroll through touch to explore 360 scene.
<i>Actors</i>	Platform User(single Device Mode)
<i>Extends</i>	None
<i>Main Success Scenario</i>	- On the home screen, user can simply click on the image and system will display the image. - User can explore the image Using touch screen scroll the image left, right, up, bottom to view.
<i>Alternate Flows</i>	- If user wants to view the image After successful completion of image capturing process, user will prompted to save/cancel the captured images. - On Save the final 360 image will be displayed on the screen.
<i>Exceptions</i>	If viewer won't be able to display the image the error will be displayed "problem is occurred, please retry"
<i>Includes</i>	

Table 8: UC-6

Use Case #7 – Type Selection

<i>UC Identifier</i>	UC-7
<i>Purpose</i>	To enable user to set devices Master device and other as participant devices.
<i>Preconditions</i>	The user must have login minimum 2 device are present.
<i>Post conditions</i>	Roles Master or Participant will be assigned to each device, and Master device is connected and synchronized with all other participant devices.
<i>Actors</i>	Platform User(Multiple Phones Camera's)
<i>Extends</i>	None
<i>Main Success Scenario</i>	- After clicking on Sync Cam user will be prompted to select their device as Master device or Participant device. - Step 1 will be performed on each device that will part of the image capturing process. - Master device will be redirect to nearby devices connection process.

	Participants devices will be in waiting state for connection with Master device.
Alternate Flows	- If no nearby devices available system display a message “No, devices found, please ensure other devices are in waiting state” - More than one device attempt to join as master device system notify user “only one Master device is allowed”.
Exceptions	A device does not support the required communication protocols, the system display an error message indicating “your device does not support the required communication protocols”.
Includes	Login

Table 9: UC-7

Use Case #8 – Nearby Devices

UC Identifier	UC-8
Purpose	To enable user to view the available nearby devices list.
Preconditions	User must have turn on their wifi-direct.
Post conditions	A list of nearby available devices will be displayed on the screen.
Actors	Platform User(Master Device)
Extends	None
Main Success Scenario	After Selecting device as Master device user get the list of available devices.
Alternate Flows	If the wifi is turned off user prompted to turn on the wifi. after turning on system will display the available devices.
Exceptions	If there is no available device is found a prompt will be display “No, available device is found, please make sure to turn on wifi on each device”
Includes	mode selection

Table 10: UC-8

Use Case #9 – Session Devices

UC Identifier	UC-9
Purpose	To enable user to select their participant devices from the available devices.
Preconditions	User must have turn on their wifi-direct.
Post conditions	A prompt will be displayed on the selected devices asking for their sequence number confirmation.
Actors	Platform User(Master Device)
Extends	None

Main Success Scenario	• A list of available participant devices will be displayed on the master device screen. • Master User select the devices that will participate in image capturing process. • And prompt will displayed on each selected device. • Participant devices tap their corresponding numbering from left to right.
Alternate Flows	• On or more devices did not respond a prompt will be displayed would you like to retry or remove unresponsive device.
Exceptions	If two devices try to choose the same number a message will display “This number is already taken, please choose another number”
Includes	Nearby devices

Table 11: UC-9

Use Case #10 – Set Timer

UC Identifier	UC-10
Purpose	To enable user select appropriate timer e.g 5sec, 10 sec.
Preconditions	All devices Master and participant devices are connected.
Post conditions	The timer will be set for appropriate time and after pressing initiate or capture master device take that selected time and start simultaneously capturing process on each device.
Actors	Platform User(Master Device)
Extends	None
Main Success Scenario	• On top right corner the timer icon is displaying user click and select appropriate timer. • On selection, system will save the selected timer and will be used for future image capturing process.
Alternate Flows	• User can select disabled timer to disable the timer and then image captured instantly.
Exceptions	If user try to change timer while image processing in progress the following prompt will display “Cannot change timer while capturing”
Includes	None

Table 12: UC-10

Use Case #11 – Assign Sequence Numbering

UC Identifier	UC-11
Purpose	To enable user to give numbering to each participant device.

Preconditions	Make sure on all participant devices the sequence number popup is being displaying.
Post conditions	A sequence number is assigned to each participant device.
Actors	Platform User(Participant Device)
Extends	None
Main Success Scenario	- Participant user select sequence number on each participant devices. - Participant devices confirm the selection and goes to waiting state for initiating process from the master device.
Alternate Flows	Master device will wait for at least one minute for confirmation from the participant device in case of no confirmation in that time frame master device asks prompt user to “continue waiting or remove unresponsive participant devices”
Exceptions	Any participant user try to assign same number to multiple phones a error message will display “ cannot assign same number to multiple devices, select available number assign left to right.
Includes	None

Table 13: UC-11

Use Case #12 – Save Image

UC Identifier	UC-12
Purpose	To enable user to save the image.
Preconditions	Image shall be captured before saving.
Post conditions	Image saved, and displays on the screen.
Actors	Platform User, Master Device
Extends	None
Main Success Scenario	- After completing image capturing session user is prompted to “Save & Cancel “buttons. - On save the system will save the image with time as title for the image.
Alternate Flows	If user clicks on the “Cancel” all the session images will be deleted from the storage.
Exceptions	If Mobile storage is full and no free space is available system will display “Your device space is full, try to free some space and try again”.
Includes	None

Table 14: UC-12

Use Case #13 –Single Device Guide

<i>UC Identifier</i>	UC-13
<i>Purpose</i>	Guide for user on how to capture 360 using single mobile.
<i>Preconditions</i>	None
<i>Post conditions</i>	Guide for Image capture using single device will be displayed.
<i>Actors</i>	Platform User
<i>Extends</i>	None
<i>Main Success Scenario</i>	- At home section user can navigate or click on the Single Device guide. - A guide on how to rotate, align mobile camera to cover entire 360-degree using single mobile camera.
<i>Alternate Flows</i>	None
<i>Exceptions</i>	None
<i>Includes</i>	None

Table 15: UC-13

Use Case #14 –Multiple Device Guide

<i>UC Identifier</i>	UC-14
<i>Purpose</i>	Guide for user on how to capture 360 using Multiple phone camera's
<i>Preconditions</i>	None
<i>Post conditions</i>	Guide for Image captures using multiple devices will be displayed on the screen.
<i>Actors</i>	Platform User
<i>Extends</i>	None
<i>Main Success Scenario</i>	- At home section user can navigate or click on the Multiple Device guide. - A guide on how to assign roles, establishing connection between phones and initiate image capture session.
<i>Alternate Flows</i>	None
<i>Exceptions</i>	None
<i>Includes</i>	None

Table 16: UC-14

Use Case #15 – Rename Image

<i>UC Identifier</i>	UC-15
<i>Purpose</i>	To rename an image.
<i>Preconditions</i>	None
<i>Post conditions</i>	The name of the image will be updated to new chosen name.
<i>Actors</i>	Platform User
<i>Extends</i>	None

Main Success Scenario	- List of previously captured images is being displaying on the screen. - User click on the top right corner three dots. - Navigate to rename a popup will display. - User can type their new desired name and click “ok”.
Alternate Flows	If user clicks “cancel” the previous name will persist.
Exceptions	If the image has been deleted while renaming “image not exists”.
Includes	None

Table 17: UC-15

Use Case #16 – Delete Image

UC Identifier	UC-16
Purpose	To delete an image.
Preconditions	None
Post conditions	The image will be deleted permanently from the phone storage.
Actors	Platform User
Extends	None
Main Success Scenario	- List of previously captured images is being displaying on the screen. - User click on the top right corner three dots. - Navigate to delete a popup will display. - Pop ask user to “are you want to delete this image”. - On yes, the image will be deleted.
Alternate Flows	If user clicks “cancel” the image will persist and will not be deleted.
Exceptions	None
Includes	None

Table 18: UC-16

Use Case #17 – Search Image

UC Identifier	UC-17
Purpose	To Search an image.
Preconditions	User must be entered at least 1 character.
Post conditions	Matching image will be displayed to the user.
Actors	Platform User
Extends	None
Main Success Scenario	- List of previously captured images is being displaying on the screen. - User can type the name of image that he/she wants to search.

	• System searches the typed name in the history of previously captured images. • System returns the image.
<i>Alternate Flows</i>	If the typed image is not present System displays “No images found matching your search”.
<i>Exceptions</i>	None
<i>Includes</i>	None

Table 19: UC-17

2.4 Use Case Diagram



Figure 1: Use case diagram

3. System Design

3.1 Design Considerations

Assumptions and Dependencies

The system has following assumptions and dependencies:

- User should thoroughly read and understood the guide to use the system to get desired results.
- The device should have necessary sensors like accelerometer and gyroscope.

Limitations

The system has following limitations:

- It could take long time to generate the result on the devices having low processing power and memory.

Risk and Solution

Risk : If there are a lot of users having the devices of low computational power and memory the system will be unable to be used for large number of users.

Solution: Develop a cloud server to create 360-degree image from input 2D images.

3.2 Requirements Traceability Matrix

Shows how each design element traces back to a specific requirement in the Software Requirements Specification (SRS), ensuring that all requirements are accounted for in the design. Use table to document requirements traceability, where requirement ID is a unique identifier assigned to the requirement, requirement description refers to brief description of the requirement discussed. Design specification entails list of the components/classes/algorithm where the requirement is addressed. ¹

Following is the requirement traceability matrix to trace each functional requirement of the system.

Table 20 Requirements Traceability Matrix

Requirement ID	Requirement Description	Design Specification
----------------	-------------------------	----------------------

FR-1	User shall be able to register using their email and password.	Component “Authentication System”
FR-2	User shall be able to log in using their credentials.	Component “Authentication System”
FR-3	User shall be able to reset their password via email verification.	Component “Authentication System”
FR-4	User shall be able to capture the image.	Class “Camera”
FR-5	System shall guide user to capture next image.	Class “OrientationMngr”
FR-6	User shall be able to save the final image.	Class “StorageMngr”
FR-7	User shall be able to select the device as “server” or “client” device.	Component “Multi-device”
FR-8	Server user shall be able to view the nearby devices.	Class “Seerver”
FR-9	Server user shall be able to select the client devices which will participate in image capturing session.	Class “Server”
FR-10	Client user shall be able to assign sequence numbering to each device.	Class “Server”
FR-11	Server user shall be able to set the timer for image capturing.	Component “Multi-device”
FR-12	Server user shall be able to initiate image capturing process.	Class “Server” Component “multi-device mode”
FR-13	User shall be able to view the final 360-degree image in an interactive manner.	Component “Viewer”
FR-14	User shall be able to view the guide for single device Image capturing.	Component “Single device”
FR-15	User shall be able to view the guide for multiple device image capturing.	Component “multi-device mode”
FR-16	User shall be able to search a previously captured image using name.	Class “StorageMngr” Component “Home module”
FR-17	User shall be able to rename the captured image.	Class “StorageMngr” Component “Home module”

FR-18	User shall be able to delete an image.	Class “StorageMngr” Component “Home module”
FR-19	User shall be able to share images.	Component “Home module”

3.3 Design Models

Following are the models and diagrams used to describe the system and its different components.

3.3.1 Architectural Design

The system is going to be created using OOP paradigm so following is the class diagram created using UML notations.

Class diagram

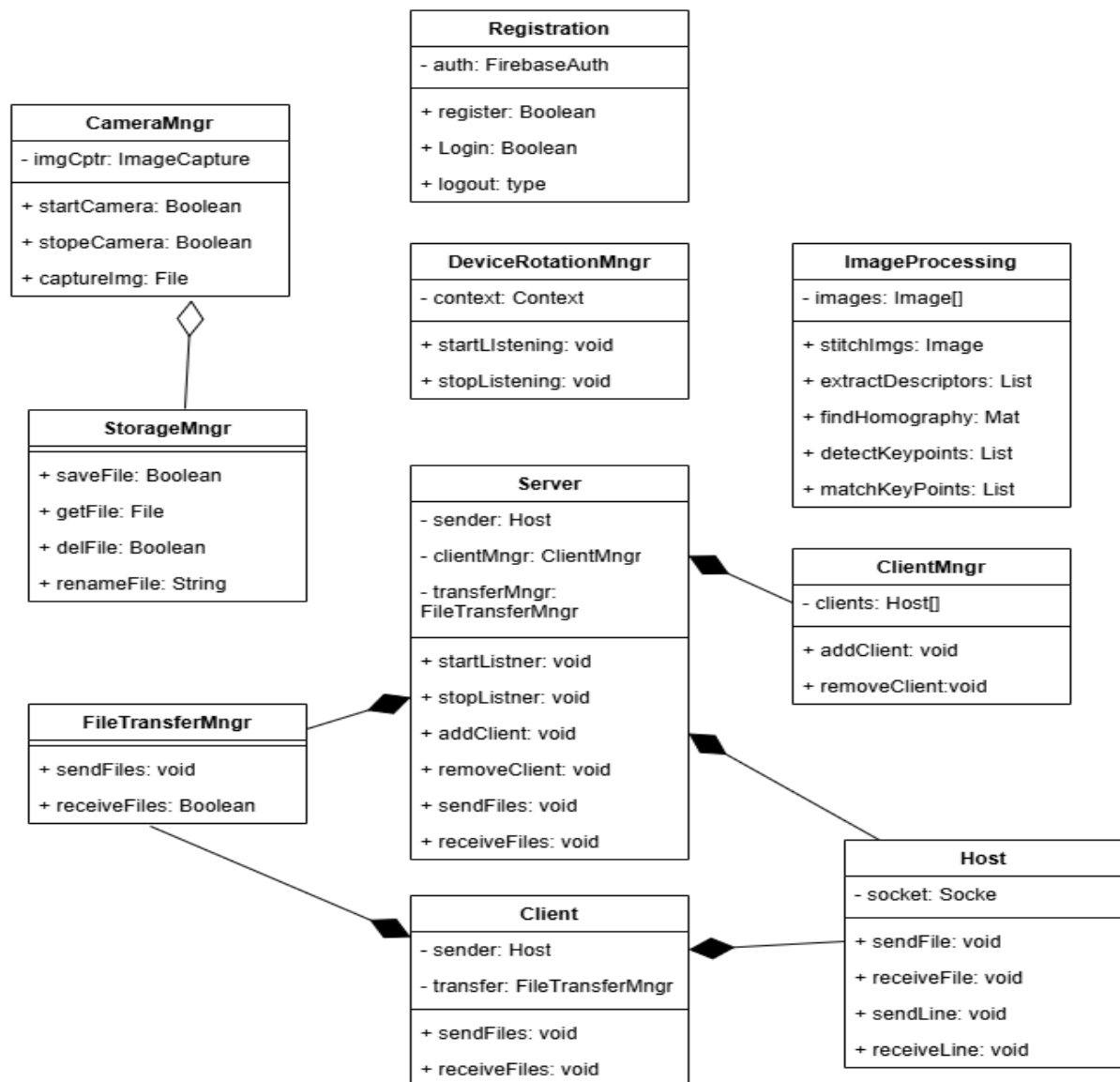


Figure 2 Class diagram

Application architecture

We are using MVVM (Model, View, View Model) architecture model to create the application. MVVM provide automatic reactive behavior for jetpack compose which is optimized to update the UI.

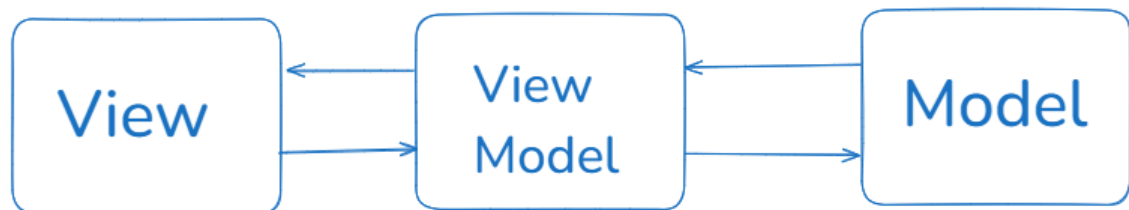


Figure 3 Application architecture

3.3.2 Data Design

The system has not complex database there are only two folders of images, one has resulted 360-degree images and other have temporarily images captured by user to create 360-degree image.

Data storage model

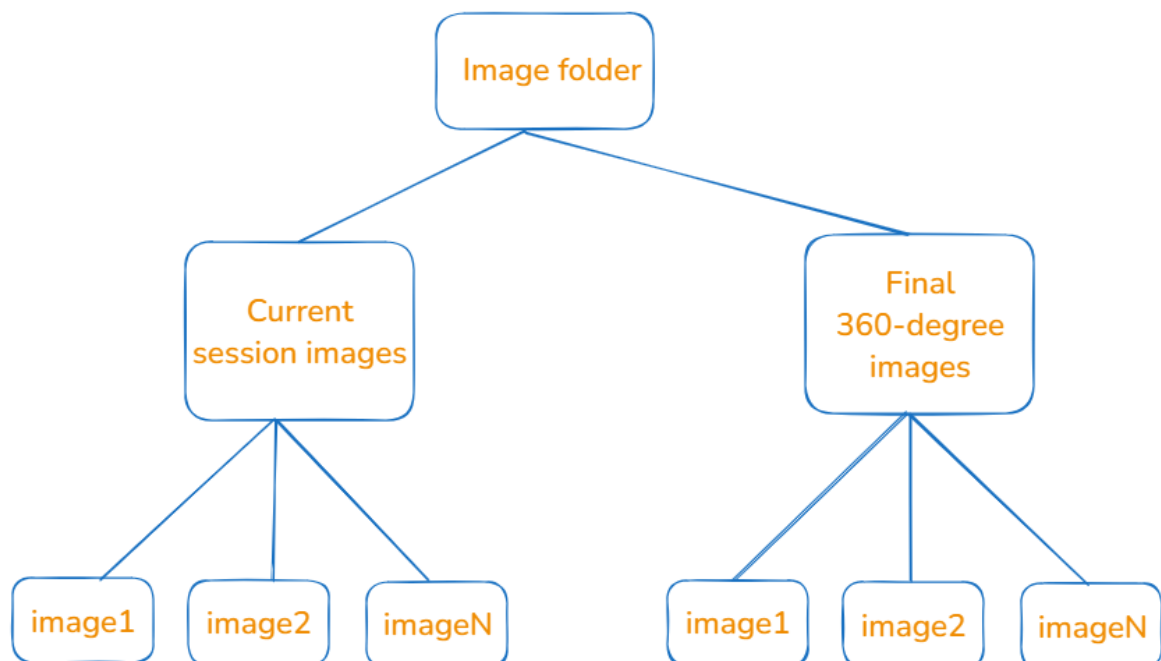


Figure 4 Database design

3.3.3 Data Dictionary

We are going to adopt the Object Oriented Programming paradigm for the business logic implementation.

List of objects

Following is the list of all objects used in the application along with the description of their attributes and methods.

Camera

Attribute name	Type	Description
imgCapture	ImageCapture	An object used to capture the image.
Method name	Parameters	Description
startCamera	Callback function	Configure camera and start the camera preview on screen.
stopCamera	Non	Stop and release the camera resources.
captureImg	callBack function	Trigger the image capturing process and invoke call back function on success full capturing.

FileManager

Attribute	Type	Description
folder	File	A complete path of folder where captured images will be stored.
Method	Parameters	Description
createFolder	Folder name	Create a folder in storage using the folder attribute.
saveImg	Image bytes Image name	Get the image bytes array and name and store the image in the folder attribute path.

deleteImg	Image name	Get the image name and delete the image from attribute folder.
renameImg	Old name New name	Get old and new name and rename the image.
getSavedImgs	Non	Return the list of all images stored in the folder attribute.

OrientationMngr

Attribute	Type	Description
referenceAngle	List<float>	Store the angles used as reference to guide the user to capture next image.
currentAngles	List<float>	Angles according to the current orientation of device.
sensor	SensorMngr	Object to get the sensor data.
Method	Parameters	Description
startOrientation	Non	Create a folder in storage using the folder attribute.
stopOrientation	Non	Get the image bytes array and name and store the image in the folder attribute path.
getAngleDiff	Non	Return the difference between reference and current angle for each axis.

SensorMngr

Attribute	Type	Description
onValuesChange	Function reference	Call back to call on the change in the sensor values.
Method	Parameters	Description
calculateAngle	Axis1 value Axis2 value	Calculate the angle in degree from given values of any two axis.
stopSensor	Non	Stop the sensor reading.

startSensor	Non	Start sensor reading.
onValuesUpdate	Non	Take action at each time the sensor data is changed.
getAngles	Non	Return the list of angles of all three axis.

CommandProcessor

Attribute	Type	Description
Method	Parameters	Description
openCamera	Devices list	Send command to open the camera on each device mentioned in provided list.
closeCamera	Devices list	Send command to open the camera on each device mentioned in provided list.
captureImage	Devices list	Send command to capture the image to each device mentioned in provided list.
sendImgTo	Device	Send image to the provided device.
sendSequenceNo	Device list	Send a pop up to each device provided in the list.

Command

Attribute	Type	Description
captureImg	Int	Capture image command enumeration
sendPopUp	Int	Send sequence pop-up command enumeration
receiveImg	Int	Receive image command enumeration
updateConnStatus	Int	Update connection status on participant device command enumeration

Method	Parameters	Description
createFolder	Folder name	Create a folder in storage using the folder attribute.
saveImg	Image bytes Image name	Get the image bytes array and name and store the image in the folder attribute path.
deleteImg	Image name	Get the image name and delete the image from attribute folder.
renameImg	Old name New name	Get old and new name and rename the image.
getSavedImgs	Non	Return the list of all images stored in the folder attribute.

3.4 Behavioral Model

To conceptualize the interaction of user throughout the system we can describe the system and user action using following high-level description.

- The user will open the application's home page. On the home page the user will be able to search images and create new one using single device mode.
- If user want to process with multi-device mode one will select multi-device mode tab and select the current device as server.
- For client devices same steps will be repeated on each device select the device as client device instead of server and perform necessary actions to connect and communicate with server device.
- On server device the user will scan and connect with participant device, send commands to capture and receive the images from connected devices after necessary image processing. After successful execution of these steps the user will be able to view and interact with resulted 360-degree image on master device.

Following is the pictorial view of high level steps performed to capture and save the image.

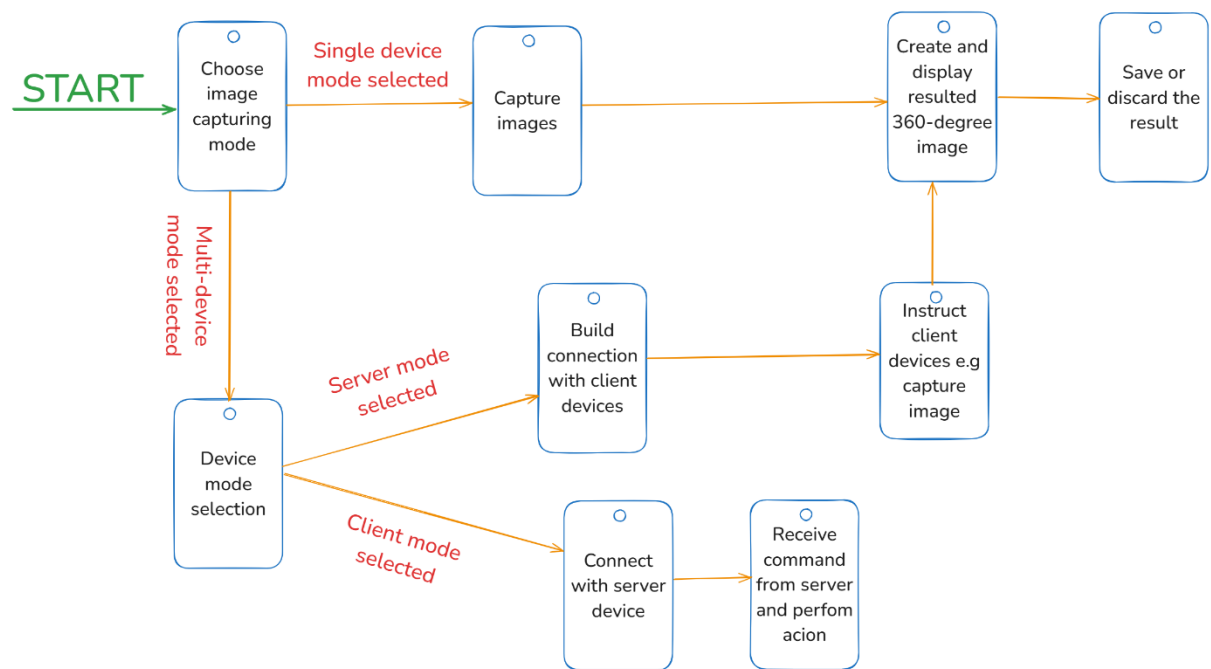


Figure 5 Flow of high level steps to capture and save the image

3.5 Design Decisions

Following are the design choices used for the system.

Application architecture

To design the system the MVVM (Model, View, View Model) architecture is chosen. MVVM is the best choice for complex applications and it also provides the automatic reactive behavior with Jetpack Compose.

Programming paradigm

To organize the code of the application the OOP paradigm is used because the Kotlin language, which is the official language for Android app development, follows the OOP paradigm like Java. Also, the OOP paradigm is the best to model system's entities. OOP provides flexibility to the code so it is the best choice to add the features in the future.

Database design

For the database, a simple file system design is used because the application does not have complex database entities and images will be used on local storage. So, the file system design is used instead of complex and advanced database models like relational database.

3.6 Summary

The vision360 have hierarchical database design because it has not any complex data to store. Although we can use relational database to manage the image names and searching but the app is managing all the images on local storage and android's built-in features related to storage are enough for the system requirements. The system is using OOP for code structure and MVVM as application architecture because they provide flexibility and automatic reactivity.

4. Implementation

This chapter explains how the main parts of the system are built. It focuses on the logic behind the features and the techniques used to make them work. It includes pseudocode for the important algorithms, shows the tools and APIs used, and provides information about the GitHub repository used for managing the project.

4.1 Algorithm

This algorithm is responsible for extracting key points and descriptors from the input images using a selected feature descriptor method such as SIFT, SURF, ORB, or BRISK. These features form the foundation for further processing, such as matching and stitching

Feature Extraction Methods
Input: image: A grayscale image
Output: keypoints, features: Detected keypoints and corresponding descriptors
1: Function select_descriptor_method(image, method): 2: If method == 'sift': descriptor ← SIFT_create() 3: Else if method == 'surf': descriptor ← SURF_create() 4: Else if method == 'orb': descriptor ← ORB_create() 5: Else if method == 'brisk': descriptor ← BRISK_create() 6: (keypoints, features) ← descriptor.detectAndCompute(image, None) 7: Return (keypoints, features)

Table 21 Algorithm 1

4.1 Algorithm 2: Create Matching Object

This algorithm creates a feature matcher object based on the type of descriptor used. It ensures that appropriate norms are applied depending on the descriptor type (e.g., L2 for SIFT/SURF and Hamming for ORB/BRISK)

Feature Extraction Methods
Input: method: Feature matching method crosscheck: Boolean flag
Output: bf: Brute-Force matcher object

```

1: Function create_matching_object(method, crossCheck):
2: If method in ['sift', 'surf']:
3: bf ← BFMatcher(NORM_L2, crossCheck)
4: Else if method in ['orb', 'brisk']:
5: bf ← BFMatcher(NORM_HAMMING, crossCheck)
6: Return bf

```

Table 22 Algorithm 2

4.1 Algorithm 3A: Keypoint Matching (Direct)

This algorithm performs direct keypoint matching using the brute-force method. It compares descriptors from two images and returns a sorted list of matches based on distance

Feature Extraction Methods

Input: features_train_img, features_query_img: Descriptors of both images
method: Feature extraction method used

Output: best_matches: List of sorted matches

```

1: Function key_points_matching(f1, f2, method):
2: bf ← create_matching_object(method, crossCheck=True)
3: matches ← bf.match(f1, f2)
4: best_matches ← sort(matches by ascending distance)
5: Return best_matches

```

Table 23 Algorithm 3

4.1 Algorithm 3B: Keypoint Matching (KNN with Ratio Test)

This algorithm enhances keypoint matching accuracy using the k-nearest neighbors approach combined with Lowe's ratio test. It filters out weak matches and retains only strong, reliable correspondences.

Feature Extraction Methods

Input:

- features_train_img, features_query_img: Descriptors of both images
- ratio: Lowe's ratio test threshold (e.g., 0.75)
- method: Feature extraction method

- **Output:** matches: Filtered list of good matches

```

1: Function key_points_matching_KNN(f1, f2, ratio, method):
2: bf ← create_matching_object(method, crossCheck=False)
3: rawMatches ← bf.knnMatch(f1, f2, k=2)
4: matches ← []
5: For each (m, n) in rawMatches:
6: If m.distance < ratio × n.distance:
7: matches.append(m)
8: Return matches (keypoints, features)

```

Table 24 Algorithm 4

4.1 Algorithm 4: Full Pipeline

This algorithm integrates all previous steps into a complete image processing pipeline. It includes image preprocessing, feature extraction, matching (both direct and KNN-based), and visualizing the matched features.

Feature Extraction Methods
Input: train.jpg, query.jpg images feature_extraction_algo (e.g., 'sift') matching_method (e.g., 'bf')
Output: result_direct, result_knn: Images showing direct and KNN matches
<pre> 1: Load train_img and query_img 2: Convert both images to grayscale → train_img_gray, query_img_gray 3: keypoints_train, features_train ← select_descriptor_method(train_img_gray, feature_extraction_algo) 4: keypoints_query, features_query ← select_descriptor_method(query_img_gray, feature_extraction_algo) 5: matches ← key_points_matching(features_train, features_query, feature_extraction_algo) 6: result_direct ← draw first 100 matches using drawMatches() 7: knn_matches ← key_points_matching_KNN(features_train, features_query, 0.75, feature_extraction_algo) 8: result_knn ← draw first 100 KNN matches using drawMatches() </pre>

Table 25 Algorithm 5

4.2 External APIs/SDKs

Describe the third-party APIs/SDKs used in the project implementation in the following table. Few examples of APIs are provided in the table.

Name of API and version	Description of API	Purpose of usage	List down the API endpoint/function/class in which it is used
OpenCV 10.4	Computer vision functions	Feature matching, warping, stitching	cv.imread(), cv.cvtColor(), cv.SIFT_create(), cv.ORB_create(), cv.BFMatcher(), cv.drawMatches()

Table 26 Details of APIs used in the project

4.3 Code Repository

In order to manage version control and ensure effective collaboration throughout the group project, **Git** has been adopted as the primary version control system. Git provides a robust and efficient way to track changes, coordinate among team members, and maintain a consistent development workflow. All project-related resources, including source code, documentation, Software Requirement Specifications (SRS), and implementation files, are stored and managed within a centralized Git repository. This approach ensures that every team member has access to the most recent version of the project and facilitates streamlined collaboration and change tracking.

Git Repository Link:

The Git repository for this project can be accessed via the following link:

<https://github.com/mehboob14/Creating-360-Degree-Images-Using-Mobile-Cameras>

4.4 Summary

This chapter explained the implementation details of the project. It described the key algorithms used for feature extraction and matching, the role of external APIs like

OpenCV, and how Git was used to manage the code. These components are essential for building a working system that meets the project goals.

5. Testing and Evaluation

The chapter include the test cases for different types of testing including unit testing, functional testing, integration testing and performance testing. For Unit testing we implemented automated test cases because it is difficult to test single method of class using manual testing. For other remaining types of testing manual test cases are written.

5.1 Unit Testing (UT)

For the unit testing automated test cases are implemented and uploaded on the following link:

<https://github.com/mehboob14/Creating-360-Degree-Images-Using-Mobile-Cameras>

5.2 Functional Testing (FT)

In this functional testing, the functionality of each of the module/ stakeholders' view is tested. This is to ensure that the system produced meets the specifications and requirements

Test Case ID	TC1
Requirement ID	FR1
Title	User Registration
Description	User registers using email and password.
Objective	To verify that the user can register successfully using email and password.
Driver / Precondition	User is on the registration screen.
Test Steps	1. Enter a valid email. 2. Enter a valid password. 3. Click on the Register button.
Input	Email: user@example.com, Password: SecurePass123

Expected Result	User is registered and redirected to the login screen.
Actual Result	User is registered and redirected to the login screen.
Test Status	Pass
Remarks	None

Table 27 TC1

Test Case ID	TC2
Requirement ID	FR2
Title	User Login
Description	User logs in using email and password.
Objective	To verify that the user can log in with correct credentials.
Driver / Precondition	User is registered and on the login screen.
Test Steps	1. Enter registered email. 2. Enter correct password. 3. Click on the Login button.
Input	Email: user@example.com, Password: SecurePass123
Expected Result	User is logged in and navigated to the home screen.
Actual Result	User is logged in and navigated to the home screen.
Test Status	Pass
Remarks	None

Table 28 TC2

Test Case ID	TC3
Requirement ID	FR3
Title	Password Reset via Email
Description	User resets password via email verification.
Objective	To verify password reset functionality through email.
Driver / Precondition	User is on the password reset screen.
Test Steps	1. Enter registered email. 2. Click on the Reset Password button. 3. Check email and follow the link. 4. Enter new password and confirm.
Input	Email: user@example.com, New Password: NewSecure123

Expected Result	User receives reset email and successfully updates password.
Actual Result	User receives reset email and successfully updates password.
Test Status	Pass
Remarks	None

Table 29 TC3

Test Case ID	TC4
Requirement ID	FR4
Title	Capture Image (Single Device)
sDescription	User captures an image using the device camera.
Objective	To verify that user can capture image in single device mode.
Driver / Precondition	User is on the capture screen in single device mode.
Test Steps	1. Open camera. 2. Capture image. 3. Display image saving message.
Input	Tap camera button.
Expected Result	Image is captured and displayed for confirmation.
Actual Result	Image is captured and displayed for confirmation.
Test Status	Pass
Remarks	None

Table 30 TC4

Test Case ID	TC5
Requirement ID	FR5
Title	Guide for Next Image
Description	System guides user to capture the next image.
Objective	To verify that system provides guidance for next image.
Driver / Precondition	User has captured first image.
Test Steps	1. Capture first image. 2. Observe guidance. 3. Follow guidance.
Input	System-generated instructions.
Expected Result	System displays instructions to capture the next image.

Actual Result	System displays instructions to capture the next image.
Test Status	Pass
Remarks	None

Table 31 TC5

Test Case ID	TC6
Requirement ID	FR6
Title	Save Final Image
Description	User saves the final stitched image.
Objective	To verify the final image can be saved successfully.
Driver / Precondition	Image capturing and stitching is completed.
Test Steps	1. Tap Save button. 2. Confirm save.
Input	Tap save and confirm.
Expected Result	Final image is saved in device storage.
Actual Result	Final image is saved in device storage.
Test Status	Pass
Remarks	None

Table 32 TC6

Test Case ID	TC7
Requirement ID	FR7
Title	Select Device Role
Description	User selects the device as server or client.
Objective	To verify that device role can be selected.
Driver / Precondition	App is launched and multi device mode is selected.
Test Steps	1. Choose device mode. 2. Connect the devices accordingly.
Input	Select 'Server' from role options.
Expected Result	Device is assigned as Server.
Actual Result	Device is assigned as Server.
Test Status	Pass
Remarks	None

Table 33 TC7

Test Case ID	TC8
Requirement ID	FR10
Title	Assign Sequence Number

Description	Server assigns sequence number to connected devices.
Objective	To verify that sequence number is assigned by the server.
Driver / Precondition	Devices are connected to server.
Test Steps	1. Server views connected devices. 2. Assigns sequence. 3. Confirms assignment.
Input	Assign 1, 2, 3 to devices.
Expected Result	Sequence numbers are saved and acknowledged by clients.
Actual Result	Sequence numbers are saved and acknowledged by clients.
Test Status	Pass
Remarks	None

Table 34 TC8

Test Case ID	TC9
Requirement ID	R11
Title	Set Timer
Description	Master user sets timer for image capturing.
Objective	To verify the timer can be configured.
Driver / Precondition	Devices are ready for capture.
Test Steps	1. Open timer settings. 2. Set desired time. 3. Save configuration.
Input	Set timer to 10 seconds.
Expected Result	Timer is set and visible on all devices.
Actual Result	Timer is set and visible on all devices.
Test Status	Pass
Remarks	None

Table 35 TC9

Test Case ID	TC10
Requirement ID	FR12
Title	Initiate Image Capture
Description	Master user initiates synchronized image capture.
Objective	To verify the capture process can be started by master user.
Driver / Precondition	Timer and roles are configured.
Test Steps	1. Tap capture button. 2. Countdown starts. 3. Images captured automatically.

Input	Tap capture button.
Expected Result	All devices capture image at the same time.
Actual Result	All devices capture image at the same time.
Test Status	Pass
Remarks	None

Table 36 TC10

Test Case ID	TC11
Requirement ID	FFR13
Title	View 360-Degree Image Interactively
Description	Open and interact with the final image.
Objective	Confirm image loads and is interactable.
Driver / Precondition	Final image is available.
Test Steps	1. Open viewer module. 2. Select a 360-degree image. 3. Swipe/rotate to interact.
Input	Gesture input.
Expected Result	Image loads and responds to interaction.
Actual Result	Image loads and responds to interaction.
Test Status	Pass
Remarks	None

Table 37 TC11

Test Case ID	TC12
Requirement ID	FR14
Title	View Guide for Single Device Capturing
Description	Access the guide for single device capture steps.
Objective	Verify guide is shown correctly.
Driver / Precondition	App is open.
Test Steps	1. Navigate to Help/Guide section. 2. Select "Single Device Guide".
Input	Click guide link.
Expected Result	Guide for single device capturing is displayed.
Actual Result	Guide is displayed properly.
Test Status	Pass
Remarks	None

Table 38 TC12

Test Case ID	TC13
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Requirement ID	FR15
Title	View Guide for Multi-Device Capturing
Description	Access the guide for multi-device capturing.
Objective	Ensure correct guide is displayed.
Driver / Precondition	App is open.
Test Steps	1. Go to Help/Guide section. 2. Select "Multi-Device Guide".
Input	Click guide link.
Expected Result	Multi-device capturing guide is displayed.
Actual Result	Multi-device capturing guide is displayed.
Test Status	Pass
Remarks	None

Table 39 TC13

Test Case ID	TC14
Requirement ID	FR16
Title	Search Image by Name
Description	Search for a saved image using its name.
Objective	Confirm image search returns correct result.
Driver / Precondition	At least one image is saved.
Test Steps	1. Go to Home screen. 2. Enter image name in search box. 3. Press Search.
Input	Image name: Test360
Expected Result	Matching image is displayed.
Actual Result	Matching image is displayed.
Test Status	Pass
Remarks	None

Table 40 TC14

Test Case ID	TC15
Requirement ID	FR17
Title	Rename Captured Image
Description	User renames a stored image file.
Objective	Ensure image is renamed correctly.
Driver / Precondition	Image is available.
Test Steps	1. Select an image. 2. Tap "Rename". 3. Enter new name. 4. Confirm rename.

Input	New name: TripView1
Expected Result	Image is renamed successfully.
Actual Result	Image is renamed successfully.
Test Status	Pass
Remarks	None

Table 41 TC15

Test Case ID	TC16
Requirement ID	FR18
Title	Delete an Image
Description	Remove a selected image from storage.
Objective	Confirm image is deleted.
Driver / Precondition	At least one image exists.
Test Steps	1. Select an image. 2. Tap "Delete". 3. Confirm deletion.
Input	Confirm delete.
Expected Result	Image is deleted and removed from list.
Actual Result	Image is deleted and removed from list.
Test Status	Pass
Remarks	None

Table 42 TC16

Test Case ID	TC17
Requirement ID	FR19
Title	Share an Image
Description	Share selected image via available apps.
Objective	Verify image can be shared.
Driver / Precondition	Image is saved.
Test Steps	1. Tap on an image. 2. Choose "Share". 3. Select sharing app.
Input	Share via: Email
Expected Result	Image is shared via selected method.
Actual Result	Image is shared via selected method.
Test Status	Pass
Remarks	None

Table 43 TC17

5.3 Integration Testing (IT)

After ensuring the individual test runs successfully, the interaction of the classes/ functions is tested to assess whether the feature is yielding the desired results.

Test Case ID	TC1
Requirement ID	FR1, FR2
Title	Authentication Workflow
Description	User completes registration and login process.
Objective	To verify registration integrates properly with login system.
Driver / Precondition	User is on the registration screen.
Test Steps	1. Register using email and password. 2. Redirect to login. 3. Log in using same credentials.
Input	Email: user@example.com Password: SecurePass123
Expected Result	User is logged in after successful registration.
Actual Result	User is logged in after successful registration.
Test Status	Pass
Remarks	None

Table 44 TC18

Test Case ID	TC2
Requirement ID	FR4, FR5, FR6
Title	Single Device Image Capture Integration
Description	Integration of image capturing, guidance, and saving functionality.
Objective	To ensure the system guides through and saves final image properly.
Driver / Precondition	User is in single device mode.
Test Steps	1. Capture image. 2. Follow guide for next. 3. Save final image.
Input	Captured image inputs.
Expected Result	Image captured through guided steps and saved.
Actual Result	Image captured through guided steps and saved.
Test Status	Pass

Remarks	None
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Table 45 TC19

Test Case ID	TC3
Requirement ID	FR7, FR10, FR11, FR12
Title	Multi Device Coordination Integration
Description	Interaction of role assignment, sequencing, timer and capture initiation.
Objective	To validate synchronization among multiple devices.
Driver / Precondition	All devices connected and assigned roles.
Test Steps	1. Assign roles. 2. Assign sequence. 3. Set timer. 4. Initiate capture.
Input	Device roles and sequence input.
Expected Result	All devices respond correctly to capture trigger.
Actual Result	All devices respond correctly to capture trigger.
Test Status	Pass
Remarks	None

Table 46 TC20

Test Case ID	TC4
Requirement ID	FR13, FR14, FR15
Title	Viewer and Guide Display Integration
Description	Integration of 360 viewer and both image capturing guides.
Objective	To verify access and rendering of viewer and guides.
Driver / Precondition	User is on viewer screen.
Test Steps	1. Open 360 image. 2. View single device guide. 3. View multi device guide.
Input	Saved 360 image.
Expected Result	Viewer and both guides display correctly.
Actual Result	Viewer and both guides display correctly.
Test Status	Pass
Remarks	None

Table 47 TC21

Test Case ID	TC5
Requirement ID	FR16, FR17, FR18, FR19
Title	Home Module Integration
Description	Interaction between search, rename, delete, and share features.
Objective	To verify user can manage captured images seamlessly.
Driver / Precondition	User has previously saved images.
Test Steps	1. Search image. 2. Rename image. 3. Delete image. 4. Share image.
Input	Image file and new name.
Expected Result	All operations execute successfully.
Actual Result	All operations execute successfully.
Test Status	Pass
Remarks	None

Table 48 TC22

5.4 Performance Testing (PT)

Test Case ID	PTC1
Requirement ID	FR2
Title	Login Response Time
Description	Measure the time taken to log in.
Objective	Ensure login process completes within acceptable time under load.
Driver / Precondition	User database populated with 1000 users.
Test Steps	1. Enter email and password. 2. Click login. 3. Measure response time.
Input	Email: testuser@example.com Password: Pass123
Expected Result	Login completes within 2 seconds.
Actual Result	Login completes within 1.8 seconds.
Test Status	Pass
Remarks	Test performed under simulated load of 100 users.

Table 49 PTC1

Test Case ID	PTC2
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Requirement ID	FR4, FR6
Title	Image Capture and Save Performance
Description	Measure time taken from image capture to save.
Objective	Validate performance of image capture and saving process.
Driver / Precondition	App is in single device mode.
Test Steps	1. Capture image. 2. Save image. 3. Measure duration.
Input	Captured image size: 5MB
Expected Result	Image is captured and saved within 3 seconds.
Actual Result	Operation completed in 2.7 seconds.
Test Status	Pass
Remarks	Measured over 10 iterations.

Table 50 PTC2

Test Case ID	PTC3
Requirement ID	FR12
Title	Multi Device Capture Synchronization Time
Description	Evaluate synchronization delay between devices.
Objective	Ensure minimal delay during synchronized capture.
Driver / Precondition	Server and 3 client devices connected.
Test Steps	1. Initiate capture from server. 2. Record timestamp on all devices.
Input	3 client devices
Expected Result	Capture starts within 200ms delay on all clients.
Actual Result	Maximum delay observed: 180ms.
Test Status	Pass
Remarks	Average delay 150ms.

Table 51 PTC3

Test Case ID	PTC4
Requirement ID	FR16
Title	Search Image Performance
Description	Measure time to search an image by name.
Objective	Validate search feature under a large dataset.

Driver / Precondition	System has 1000 saved images.
Test Steps	1. Enter image name in search bar. 2. Initiate search. 3. Record response time.
Input	Image name: Panorama_001
Expected Result	Search completes within 1 second.
Actual Result	Search completed in 0.9 seconds.
Test Status	Pass
Remarks	Search optimized using indexing.

Table 52 PTC4

Test Case ID	PTC5
Requirement ID	FR13
Title	360 Viewer Load Time
Description	Measure time taken to load the 360-degree viewer.
Objective	Ensure viewer loads efficiently with high-resolution images.
Driver / Precondition	User opens a 360 image from gallery.
Test Steps	1. Tap on image. 2. Start timer until viewer is interactive.
Input	Image size: 10MB
Expected Result	Viewer becomes interactive within 2.5 seconds.
Actual Result	Viewer became interactive in 2.3 seconds.
Test Status	Pass
Remarks	Optimized image loading confirmed.

Table 53 PTC5

Test Case ID	PTC6
Requirement ID	FR12
Title	Stitch and Generate Final Result from Input Images
Description	The system processes and stitches multiple input images to generate a final output image.
Objective	To verify the system performance under image stitching operations.
Driver / Precondition	More than one valid input images are captured and ready for processing.

Test Steps	1. Load a set of high-resolution input images. 2. Initiate the stitching process. 3. Wait for the final result generation and note the time taken.
Input	4 high-resolution images (e.g., 4K)
Expected Result	Final stitched image is generated within an acceptable time frame (e.g., ≤ 5 seconds).
Actual Result	Final stitched image generated in 4.3 seconds.
Test Status	Pass
Remarks	System handled stitching efficiently under expected input load.

Table 54 PTC6

Test Case ID	PTC7
Requirement ID	FR12
Title	Sharing the Image from One Device to Another
Description	An image file is shared from one device to another over a local connection.
Objective	To evaluate transfer speed and responsiveness during image sharing.
Driver / Precondition	Both devices are connected and ready for communication.
Test Steps	1. Initiate the capturing process from server device. 2. Note the time taken to received the image on server device.
Input	Image size: 10MB
Expected Result	Image transfer completes within an acceptable time frame (e.g., ≤ 3 seconds).
Actual Result	Transfer completed in 2.6 seconds.
Test Status	Pass
Remarks	Transfer speed was within acceptable limits.

Table 55 PTC7

5.5 Summary

This chapter covered key testing strategies applied during the project, including manual test cases for integration, functional, and performance testing. Additionally, automated unit testing was conducted, with the relevant test suite provided via an external link. These testing efforts were crucial in verifying system functionality, reliability, and performance, thereby supporting the overall project objectives of delivering a robust and well-tested software solution.

6. System Conversion

Vision360 is transitioning from an idea-phase prototype to an operational mobile application. While no legacy software is being replaced, this section outlines how the system will be deployed, tested, and maintained as it becomes fully functional.

6.1 Conversion Method

For this Android application, we have selected the **Phased Conversion** method.

This means the application will be implemented and tested in small parts or phases. As each feature is completed, it will be tested independently before integrating it into the full application. This method allows us to identify bugs or usability issues early and make necessary improvements without affecting the entire system.

Planned phases:

- **Phase 1:** Design and test core UI components (e.g., Image stitching, Communication).
- **Phase 2:** Implement and test individual features like the calculator logic and result display.
- **Phase 3:** Integrate all features and test the complete user flow.
- **Phase 4:** Final app testing, packaging.

6.2 Deployment

Deployment of our Android application will involve making it ready for real-world usage on Android devices. The deployment steps include:

1. Building the APK:

- a. Use Android Studio to build the final release version of the APK.

2. Testing on Devices:

- a. Run and test the app on both emulator and physical Android devices to ensure it works as expected.

3. Generating Signed APK (Optional):

- a. Generate a signed APK or AAB (Android App Bundle) for publishing or distribution.

4. Installation:

- a. Manually install the APK on target Android devices for testing or usage

6.2.1 Data Conversion

Since our application is being developed from scratch and does not rely on an old system or existing data, no complex data conversion is required.

There is no legacy system; however, image data will follow this structure:

- Capture images → Validate format → Store in local memory and Firebase Cloud Storage.
- For backup (Optional): Daily automatic Firebase backups will be scheduled.

6.2.2 Training

Training will be conducted via a basic user manual covering:

- **Use Case 1:** Capturing a 360° photo using a single device.
- **Use Case 2:** Capturing a 360° photo using multiple synchronized devices.

Each manual will include screenshots and step-by-step instructions.

6.3 Post Deployment Testing

Post-deployment, the following actions will be taken:

- Functional testing on multiple Android devices.
- Connectivity testing for multi-device capture.
- Performance testing for image stitching and panorama rendering.
- Edge case testing like failed connections or interrupted captures.

Steps required to assess that software is properly deployed and converted. This includes running the testcases to ensure correct conversion of system

6.4 Challenges

Challenges Encountered (so far & anticipated):

- Communication challenges

- Image stitching works as expected on Desktop, but when it transitions into kotlin its overall image quality effects.
- Lack of build in OpenCV methods to stitch vertical images.
- Distributing load between devices.

6.5 Summary

The Vision360 deployment strategy emphasizes k mitigation via a pilot approach, and Firebase integration. While the system is in development, deployment planning has addressed potential synchronization, storage, and connectivity issues, ensuring a smoother transition to operational use.

7. Conclusion

This chapter summarizes the current development status of the **Vision360** system — a mobile-based solution designed to create 360-degree images using single or multiple smartphones. It revisits the initial objectives, evaluates progress so far, presents traceability of requirements, and outlines future work needed to complete the project.

7.1 Evaluation

The following table evaluates whether the key objectives defined in Chapter 2 of the SRS have been addressed during the current development phase. Since the system is still under development, many features are partially implemented or pending.

Objectives	Status
Support image capture using single device	Partially Done
Support image capture using multiple devices	In Progress
Real-time communication between master and participant devices	Almost Completed
Guide user during image capturing process	Not Started
Enable assigning sequence numbers to devices	Not Started
Timer setting before capturing	Not Started
Allow saving, renaming, and deleting images	Partially Done
Provide an interactive 360-degree viewer	Pending
Enable user authentication (Register/Login)	Completed
Display step-by-step capturing guide (single/multi-device)	Partially Done

Table 56 Evaluation matrix

7.2 Traceability Matrix

The table below maps SRS requirements to their corresponding design elements, code modules (if any), and testing status. As development is ongoing, several modules are not yet fully implemented or tested.

Requirement ID	Requirement Description	Design Specification	Code Module	Test ID
FR-1	Register Account	Auth Module	Register.kt	Not Tested
FR-2	Login	Auth Module	Login.kt	Not Tested
FR-3	Forgot Password	Auth Module	ForgotPassword.kt	Not Tested
FR-4	Capture Image (Single Device)	Capture Module	Capture.kt	In Progress
FR-5	User Guide During Capture	Guide Overlay System	GuideOverlay.kt	Not Tested
FR-6	Save Image	Storage Manager	Not Implemented	N/A
FR-7	Master/Participant Role Selection	Role Selection Screen	Not Implemented	N/A
FR-8	View Nearby Devices	Wi-Fi Direct Handler	Not Implemented	N/A
FR-9	Select Participant Devices	Session Management Module	Not Implemented	N/A
FR-10	Assign Sequence Numbers	Sequence Manager	Not Implemented	N/A
FR-11	Set Timer	Timer Controller	Not Implemented	N/A
FR-12	Initiate Capture in Multi-device Mode	Capture Controller	Not Implemented	N/A
FR-13	View 360 Image	Image Viewer	Not Implemented	N/A

FR-14	Guide for Single Device	Guide Manager	SingleGuide.kt	Not Tested
FR-15	Guide for Multiple Devices	Guide Manager	Not Implemented	N/A
FR-16	Search Captured Images	Image List Module	Not Implemented	N/A
FR-17	Rename Captured Image	Image Manager	Not Implemented	N/A
FR-18	Delete Image	Image Manager	Not Implemented	N/A
FR-19	Share Image	Sharing Module	Not Implemented	N/A

Table 57 Traceability Matrix

7.3 Conclusion

At the current development stage, Vision360 has made progress on core modules such as user authentication and basic image capture in single-device mode. However, many major features, including the complete multi-device coordination, image sharing, interactive viewer, and full image management, are still under construction. This documentation reflects a realistic picture of work done so far and forms a baseline for tracking what remains.

7.4 Future Work

The Vision360 platform introduces the foundation for mobile-based 360-degree image capturing using single and multiple devices. However, several areas remain open for enhancement and extension by future developers or research teams:

- Advanced stitching algorithms
- Cloud-based processing
- Live session preview
- AR-based alignment guidance
- Integration with third-party platforms
- Cross-platform support